

# BrainSTEM Alliance

ChemiCutes Game

Character bios

# Carbo / Carbon

Why do kids love Carbo? Because this carbon character shows how one element can appear in diamonds, graphite, and living things!

ChemiCutes turns real science into an engaging card game that helps students understand molecular structures, adaptability, and the building blocks of life.

# Ferron / Iron

Ferron brings iron to life!

This strong, magnetic character helps kids understand how iron is used in steel, construction, and even inside our bodies.

Iron is part of hemoglobin, which helps carry oxygen through the blood. Chemicutes turns the periodic table into an adventure students can remember.

# Hydro (The Flying Spark)

**Element:** Hydrogen (H, Atomic Number 1)

**Personality:** Bouncy, impatient, and wildly energetic. He can never sit still and loves being the first to arrive anywhere.

**Special Ability:** Cosmic Floating. He can lift anything into the air and zoom through space faster than a rocket.

**Fun-Fact:** He is the lightest and oldest element in the entire universe, making up most of the stars!

Trading cards

# FERRON

THE MAGNETIC LEADER

26  
ATOMIC  
NUMBER

**STRONG**  
Durable & dependable

**MAGNETIC**  
Attracts & controls metallic objects

**Fe**

**IRON**



## FERRON

**IRON**

Symbol: Fe | Atomic Number: 26

**PERSONALITY**

- Strong, reliable
- Natural leader

**SPECIAL ABILITY**

- Attracts & controls metallic objects

**FUN FACT**

- Gives blood its red color
- Helps carry oxygen

**IRON IS ESSENTIAL!**

- IN OUR BLOOD
- IN OUR MUSCLES
- IN OUR IMMUNE SYSTEM
- IN MACHINES & BUILDINGS
- IN MAGNETS & TOOLS

**RARITY ON EARTH**

Very Abundant

★★★★★

**COMMON FORMS OF IRON**






IRON ORE   STEEL   CAST IRON   WROUGHT IRON

26

**Fe**

THE ELEMENT THAT KEEPS THE WORLD TOGETHER





# HYDRA

THE ULTIMATE CONNECTOR



1

ATOMIC NUMBER

POWERFUL & LIGHT

Lightest element in the universe!

CLEAN ENERGY

Hydrogen fuel powers the future!

ULTIMATE CONNECTOR

Joins with almost anything to create new compounds!

H

HYDROGEN



## HYDRA

### HYDROGEN

Symbol: H | Atomic Number: 1

**PERSONALITY**

- Energetic & curious
- Loves adventure & teamwork

**SPECIAL ABILITY**

- Can combine with almost anything
- Creates new compounds

**FUN FACT**

- Powers rockets
- Could be clean fuel of the future

**HYDROGEN POWERS THE FUTURE!**

- ROCKET FUEL
- CLEAN ENERGY
- GREEN TECHNOLOGY
- FUEL CELLS
- WATER PRODUCTION

**RARITY ON EARTH**

Very Abundant

★★★★★

**HYDROGEN IN ACTION**

WATER (H<sub>2</sub>O)   AMMONIA (NH<sub>3</sub>)   METHANE (CH<sub>4</sub>)   ROCKET FUEL   FUEL CELL

1

THE ELEMENT THAT STARTS IT ALL

H

Fundraising pitch

What if learning chemistry felt like collecting heroes, battling friends, and discovering real scientific powers? **ChemiCutes** is an educational card game that transforms the periodic table into an exciting adventure for children aged 8-14.

Every character is based on a real chemical element, with abilities inspired by actual science, making learning both fun and memorable.

Created by **BrainSTEM Alliance**, a Calgary non-profit dedicated to making STEM education accessible and engaging, **ChemiCutes** helps young learners build curiosity, critical thinking, and confidence in science through play.

Beyond the game itself, every purchase supports free STEM programs that reach students in local schools and communities.

We are currently in the prototype stage and need your support to bring **ChemiCutes** to classrooms, homes, and game tables everywhere. By backing this campaign, you're helping inspire the next generation of scientists, engineers, and innovators.

Join us today and help turn chemistry into an adventure that every child can enjoy!

Social Media posts





# MEET THE CHEMICUTES!

Hydra ⚡ Hydrogen      Carbo 💎 Carbon      Ferron ⚡ Iron

**Tiny element heroes. Real science powers. Collect. Play. Learn.**

#ChemiCutes      #CollectThemAll      #STEMFun



One thing AI got wrong

When the prompt did not have a specific role, the output was plain and simple.

When we added specifics to the role like "You are a Nobel prize in chemistry and a Pulitzer prize novel writer" the output was extremely elaborate.

The output was fixed by refining the role. The Nobel prize in chemistry was removed. Sci-Fi writer and kids-book writer was added.

The differences can be seen between Carbo, Ferron characters VS Hydro character